

Containment Without Competence: Evaluating Validation Pipelines for Language-Based Multi-UAV Planning

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Abstract: Language-based robot planners can produce fluent plans despite missing evidence or conflicting resources. We compare four validation-pipeline configurations on 1,420 controlled derivatives of 284 held-out MultiUAV-Plat tasks using frozen Qwen2.5-3B and 7B checkpoints. Stage-wise validation eliminated unsupported continuation on all 568 non-executable cases, but failed every one of 852 executable cases. A zero-call policy that always blocks achieved 20.0% strict success, exceeding the 7B stage-wise configuration’s 16.1%, showing that containment alone does not establish useful discrimination. A post-hoc audit also found that the frozen validator rejected 43 of 249 fully instantiated upstream reference plans. A paired replay that held prompts, generations, validators, and call count fixed found no placement-only change in containment, false non-execution, or static plan fidelity. The study is therefore a negative result: the tested system contains failures but does not demonstrate executable planning, and changing enforcement order alone did not reproduce the original configuration difference in the retained traces.

Keywords: physical AI safety, robot planning, failure containment, evaluation

1 Introduction

An open-ended language interface can hide a consequential physical-AI failure: a response may appear executable while relying on evidence that is absent from the robot’s visible context. Prior systems combine language models with supervision, structured planning, and motion constraints [1, 2, 3, 4]. These systems motivate modular safety checks, but they do not resolve how to evaluate a planner that avoids unsafe continuation by refusing useful requests.

We study this distinction using controlled MultiUAV-Plat derivatives [5]. Here, *containment* means that a pipeline does not advance on a registered non-executable case. *Static plan fidelity* means conformance with frozen schema, endpoint, parameter, and command validators; it does not mean simulator or physical success. Our contributions are: (1) a paired five-variant benchmark for evidence availability and resource conflict; (2) a comparison of four complete validation pipelines with frozen model and prompt versions; and (3) a refusal-aware analysis that separates containment from executable competence, audits the validator against upstream plans, and replays identical traces under early versus deferred enforcement.

2 Experimental Design

Cases. The pinned source contains 1,500 tasks in 75 sessions. Deterministic duplicate and leakage checks retained 1,473 source tasks, split by session into 899 train, 290 calibration, and 284 test clusters. Each held-out cluster yields five linked cases: canonical execute, official-alias execute, missing-evidence clarify, restored-evidence execute, and resource-conflict block. The test matrix

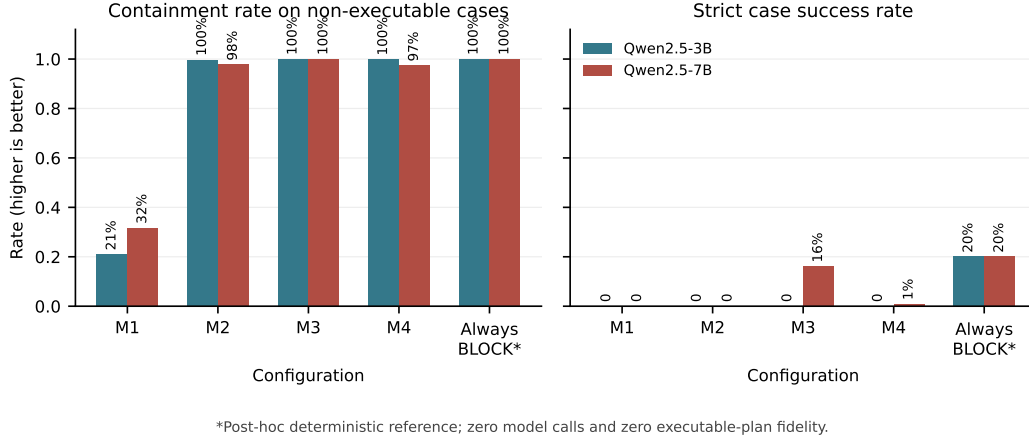


Figure 1: Containment and strict success, both higher-is-better. Always- BLOCK is a post-hoc zero-call reference with zero executable-plan fidelity.

contains 1,420 cases per method and model: 852 executable and 568 non-executable. A stratified 30-cluster pilot checked construction quality; it was not a complete human audit.

Configurations. M1 is a one-call monolithic planner. M2 adds a deterministic post-plan gate. M3 builds and checks an evidence ledger before planning and validates provenance afterward. M4 is a two-call post-plan configuration that is *model-call-count-matched* to M3. The comparison does not isolate placement: M3 and M4 also differ in prompts, validation behavior, generated content, and token counts. We therefore compare complete pipeline configurations rather than claim a causal placement effect.

Models and outcomes. We use immutable Qwen2.5-3B-Instruct and Qwen2.5-7B-Instruct revisions with deterministic decoding. Registered outcomes are unsupported continuation on the 568 non-executable cases and strict success over all 1,420 cases. Executable success requires the correct disposition and a non-empty plan passing every static validator; non-executable success requires the exact clarify or block disposition without a prohibited plan. Registered paired differences are aggregated by source task and use 10,000 fixed-seed percentile-bootstrap draws.

Post-hoc reviewer diagnostics. We score a deterministic always-BLOCK reference over unique cases, repeat paired contrasts by resampling the 15 held-out sessions, partition every executable failure into one observed location, and pass fully instantiated upstream reference API plans through the frozen validator. We also replay each retained two-call M3 trace under early and deferred enforcement while holding the prompts, both generations, validators, and call count fixed. Unresolved upstream templates are excluded rather than imputed. These diagnostics do not change the registered analysis and require no new model calls.

3 Results

Figure 1 separates two positively oriented outcomes. For 3B, containment rates for M1–M4 were 21.0%, 99.6%, 100%, and 99.8%; all strict-success rates were zero. For 7B, containment was 31.7%, 97.7%, 100%, and 97.4%, while strict success was 0%, 0%, 16.1%, and 0.8%. Every nonzero strict success was a correct non-executable outcome. All eight model–method conditions had 0/852 executable static plan fidelity.

Always-BLOCK contained every non-executable case and correctly handled the 284 conflict cases, producing 284/1,420 (20.0%) strict success. It exceeded 7B M3’s 228/1,420 (16.1%) while falsely

Table 1: Paired M3 trace replay. Unsupported continuation is over 568 non-executable cases; false non-execution and static plan fidelity are over 852 executable cases; strict success is over all 1,420 cases.

Model	Enforcement	Unsupported	False nonexec.	Static fidelity	Strict success
3B	Early	0	852	0	0
3B	Deferred	0	852	0	0
7B	Early	0	852	0	0
7B	Deferred	0	852	0	228

blocking all 852 executable and all 284 clarification cases. Thus, perfect containment and the aggregate strict-success metric do not demonstrate useful discrimination.

The registered 7B M3 minus M1 differences were -0.6831 for unsupported continuation (95% interval $[-0.7324, -0.6338]$) and 0.1606 for strict success ($[0.1507, 0.1697]$). Session-clustered sensitivity preserved the directions: -0.6829 ($[-0.7220, -0.6373]$) and 0.1605 ($[0.1513, 0.1688]$), respectively. Only 15 sessions were available, so this is a robustness check rather than a rich hierarchical analysis.

Placement-only replay. Across both checkpoints, early and deferred enforcement produced identical unsupported continuation (0/568), false non-execution (852/852), and executable static plan fidelity (0/852). For 3B, both policies also had zero strict success. For 7B, deferred enforcement preserved 228 correct final model refusals or clarifications (16.1%), whereas the early-enforcement policy recorded none as exact decisions. Under the frozen scoring contract, deterministic-gate outcomes record containment locations rather than exact CLARIFY or BLOCK decisions. The early-minus-deferred difference was -0.1606 (95% source-task-clustered interval $[-0.1697, -0.1507]$). Thus, enforcement timing alone altered neither unsupported continuation nor executable fidelity; M3’s nonzero 7B success came from the second-call decision. Because both calls were retained, this replay makes no compute-saving claim.

Where M3 executable cases failed. For 3B M3, 544/852 executable cases stopped before planning and 308/852 failed output parsing. For 7B M3, 637/852 produced a model non-execution decision, 187/852 failed output parsing, and 28/852 stopped before planning. These mutually exclusive counts show complete failure on executable cases through preplanning stops, non-execution decisions, or parse errors rather than successful executable planning.

Validator audit. Of 284 upstream reference plans, 249 were fully instantiated and 35 contained unresolved or missing template fields. The frozen validator accepted 206/249 (82.7%) and rejected 43/249 (17.3%): one at endpoint schema, three at parameter grounding, and 39 at safety bounds. Validator mismatch therefore is a plausible contributor to the zero-fidelity result, although it does not explain every model failure.

Compute. A separate exploratory campaign measured 30 clusters, all five variants, four methods, two models, and three repetitions on one RTX 3090. Relative to M1, M3 increased 3B duration by 8.86–9.49 s and board energy by 1,565–1,681 J, but reduced 7B duration by 6.23–6.82 s and board energy by 1,153–1,245 J. Relative to model-call-count-matched M4, 7B M3 generated 446.82 fewer output tokens and used 5,669–6,279 J less board energy. These differences are confounded by termination behavior and generated length; they are not evidence that stage-wise validation is intrinsically cheaper.

4 Discussion and Limitations

The central result is containment without competence. M3 suppresses unsupported continuation, but an indiscriminate block policy is stronger on the aggregate success metric and no tested method

produces a valid executable plan. The upstream-reference-plan audit additionally shows that the evaluation mechanism itself rejects a nontrivial fraction of upstream plans. The controlled replay further shows that changing deterministic enforcement order alone did not alter unsupported continuation or executable fidelity. A stronger follow-up must first reconcile and re-audit validator rules, prospectively test matched pipelines beyond one retained transcript family, and evaluate execution in an official environment.

The study uses controlled derivatives rather than live operator missions, two sizes from one model family, a 30-cluster human pilot, and one GPU for resource measurement. It performs no official simulator or physical execution. The source-task bootstrap is registered; the session bootstrap is post-hoc and has only 15 units. The resource campaign has three repetitions, and board energy is not whole-system or UAV energy. These limits prevent claims of general robot safety, practical mission capability, or broad causal superiority of one validation placement. The placement replay is conditional on retained M3 traces and is not a prospective compute experiment.

5 Conclusion

Validation-pipeline configurations can change where failures are contained, but containment is not evidence of planning competence. In this controlled study, stage-wise validation eliminated unsupported continuation while failing every executable case, underperforming always-BLOCK on aggregate strict success, and interacting with a validator that rejected 17.3% of auditable upstream plans. In replay of identical retained traces, enforcement timing changed neither unsupported continuation nor executable fidelity. The defensible contribution is a refusal-aware evaluation protocol and a negative empirical result that identifies what must be fixed before stronger physical-AI safety claims are possible.

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